

The INA 219 communicates over there I2C Address at 0x40

```
pi@raspberrypi: ~/CENG317
File Edit Tabs Help
pi@raspberrypi:~ $ cd CENG317
pi@raspberrypi:~/CENG317 $ sudo i2cdetect -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  1c  --  --  --  --
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  39  --  --  --  --  --  --  --
40: 40:  --  --  --  --  --  --  UU  --  --  --  --  --  --  --  --
50:  --  --  --  --  --  --  --  --  --  --  --  5c  --  --  5f
60:  --  --  --  --  --  --  --  --  --  --  --  6a  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
pi@raspberrypi:~/CENG317 $
```



The picture of the sensor being mounted to the case

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The INA 219 sensor that is a high-side DC current, voltage, and power monitor with an I2C-compatible interface. It is designed to precisely measure electrical parameters in a circuit. It calculates power, voltage and current. It is Bi-Directional, so it can read negative volts and amps.

```
https://life-environmentalsolution-default-rtdb.firebaseio.com

reference url
Read-only and non-realtime mode activated in the data viewer to improve browser performance
Select a key with fewer records to edit or view in realtime

ina219_readings
  -0evZCFja-5s6l33Lf2J
    current_ma: -0.2
    power_w: 0
    shunt_v: -0.00002
    timestamp: 1764083421.9837627
    vin_minus_v: 0.872
    vin_plus_v: 0.87198
```